

# PAYROLL & INVOICE CALCULATION SPECIFICATION

## Project Name

WorkForce360

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## Document Purpose

This document defines all payroll, attendance, overtime, allowance, statutory deduction, employer contribution, and invoice calculation rules used in WorkForce360.

All payroll calculations, payslip generation, invoice generation, reports, APIs, database calculations, and frontend displays must strictly follow this specification.

This document supersedes any assumptions made during development.

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## SECTION 1: SALARY STRUCTURE CONFIGURATION

### Employee Salary Components

Each employee shall have the following salary structure:

| Component             | Description                                   |
|-----------------------|-----------------------------------------------|
| Fixed Gross Salary    | Monthly gross salary agreed during onboarding |
| Tea Allowance         | Fixed monthly tea allowance                   |
| Basic Percentage      | Percentage allocation of remaining gross      |
| HRA Percentage        | Percentage allocation of remaining gross      |
| Conveyance Percentage | Percentage allocation of remaining gross      |

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### Percentage Validation Rule

The system must enforce:

Basic % + HRA % + Conveyance % = 100%

If total percentage is not equal to 100%, salary structure cannot be saved.

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## Default Percentages

System Default:

Basic = 40%  
HRA = 30%  
Conveyance = 30%

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## Employee-Level Override

Admin may modify percentages per employee.

Example:

Employee A

Basic = 50%  
HRA = 25%  
Conveyance = 25%

Employee B

Basic = 40%  
HRA = 30%  
Conveyance = 30%

---

## SECTION 2: FIXED SALARY BREAKUP

### Step 1

Calculate Remaining Gross

Formula:

```
Remaining Gross
=
Fixed Gross
-
Tea Allowance
```

Example:

```
Fixed Gross = ₹20,000

Tea Allowance = ₹500

Remaining Gross
=
20,000 - 500
=
₹19,500
```

---

## Step 2

Calculate Fixed Basic

Formula:

```
Fixed Basic
=
Remaining Gross × Basic %
```

---

## Step 3

Calculate Fixed HRA

Formula:

Fixed HRA  
=  
Remaining Gross × HRA %

## Step 4

Calculate Fixed Conveyance

Formula:

Fixed Conveyance  
=  
Remaining Gross × Conveyance %

## Validation

Fixed Basic  
+  
Fixed HRA  
+  
Fixed Conveyance  
+  
Tea Allowance  
  
=  
Fixed Gross

Must always be true.

## SECTION 3: ATTENDANCE RULES

### Attendance Status Values

| Status   | Attendance Value |
|----------|------------------|
| Present  | 1.0              |
| Half Day | 0.5              |

| Status | Attendance Value |
|--------|------------------|
| Absent | 0.0              |

## Net Payable Days

Formula:

```

Net Payable Days
=
Present Days
+
(Half Days × 0.5)

```

Example:

```

20 Present
4 Half Day
6 Absent

Net Payable Days
=
20 + (4 × 0.5)

=
22 Days

```

## Month Days

System shall use actual calendar days.

Examples:

```

January = 31

February = 28/29

March = 31

```

April = 30

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## Attendance Edit Rule

Before Payroll Generation:

Editable

After Payroll Generation:

Locked

No modification allowed.

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## SECTION 4: EARNED SALARY CALCULATIONS

### Earned Basic

Formula:

Earned Basic  
=  
(Fixed Basic × Net Payable Days)  
÷ Total Days In Month

---

### Earned HRA

Formula:

Earned HRA  
=  
(Fixed HRA × Net Payable Days)  
÷ Total Days In Month

---

## Earned Conveyance

Formula:

$$\begin{aligned} &\text{Earned Conveyance} \\ &= \\ &(\text{Fixed Conveyance} \times \text{Net Payable Days}) \\ &\div \text{Total Days In Month} \end{aligned}$$

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## Earned Tea Allowance

Formula:

$$\begin{aligned} &\text{Earned Tea Allowance} \\ &= \\ &(\text{Fixed Tea Allowance} \times \text{Net Payable Days}) \\ &\div \text{Total Days In Month} \end{aligned}$$

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## Half-Day Treatment

Half-day attendance earns:

50% Salary  
50% Tea Allowance

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# SECTION 5: OVERTIME RULES

## Allowed Values

Allowed:

1  
1.5  
2  
2.5  
3

3.5  
...

Not Allowed:

1.25  
1.75  
2.30

System shall only accept increments of:

0.5 Hour

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## Maximum OT Limit

Default:

8 Hours Per Day

Stored in Settings.

Admin may modify maximum limit.

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## Overtime Hourly Rate

Formula:

OT Rate  
=  
 $((\text{Fixed Basic} \div 30) \div 8) \times 2$

---

## Overtime Wages

Formula:



$$\begin{aligned} &\text{OT Wages} \\ &= \\ &\text{OT Rate} \times \text{Total OT Hours} \end{aligned}$$

---

## SECTION 6: SPECIAL ALLOWANCE RULES

### Special Allowance Hourly Rate

Formula:

$$\begin{aligned} &\text{Special Allowance Rate} \\ &= \\ &((\text{Fixed Basic} \div 30) \div 8) \end{aligned}$$

### Special Allowance

Formula:

$$\begin{aligned} &\text{Special Allowance} \\ &= \\ &\text{Special Allowance Rate} \times \text{Total OT Hours} \end{aligned}$$

---

## SECTION 7: GROSS PAY CALCULATION

### Earned Gross

Formula:

$$\begin{aligned} &\text{Earned Gross} \\ &= \\ &\text{Earned Basic} \\ &+ \\ &\text{Earned HRA} \\ &+ \\ &\text{Earned Conveyance} \end{aligned}$$

+  
Earned Tea Allowance

---

## Gross Pay

Formula:

Gross Pay  
=  
Earned Gross  
+  
OT Wages  
+  
Special Allowance

---

## SECTION 8: PF CALCULATIONS

### PF Eligibility

Applicable to:

All Employees

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PF Number is mandatory during onboarding.

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### Employee PF Deduction

Formula:

PF  
=  
12% × Earned Basic

---

## Employer PF Contribution

Formula:

Employer PF  
=  
 $12\% \times \text{Earned Basic}$

---

## Employer PF Admin Charges

Formula:

Employer PF Admin Charges  
=  
 $1\% \times \text{Earned Basic}$

---

# SECTION 9: ESIC CALCULATIONS

## Eligibility Rule

Employee is eligible only if:

$\text{Fixed Gross} \leq ₹21,000$

Eligibility is checked using:

Fixed Gross

Not Earned Gross.

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## Employee ESIC Deduction

Formula:

Employee ESIC  
=  
 $0.75\% \times \text{Gross Pay}$

---

## Employer ESIC Contribution

Formula:

Employer ESIC  
=  
 $3.25\% \times \text{Gross Pay}$

---

## Not Eligible

If:

Fixed Gross > ₹21,000

Then:

Employee ESIC = 0  
  
Employer ESIC = 0

---

# SECTION 10: PROFESSIONAL TAX

## Default Slabs

### Slab 1

Gross Pay  $\leq$  ₹15,000  
  
PT = ₹0

---

## Slab 2

₹15,001 - ₹20,000

PT = ₹150

## Slab 3

Gross Pay > ₹20,000

PT = ₹200

## Configuration Rule

Professional Tax slabs shall be editable through Settings.

Changes shall apply only to future payrolls.

# SECTION 11: NET SALARY CALCULATION

## Formula

```
Net Salary
=
Gross Pay
-
PF
-
ESIC
-
Professional Tax
```

## Important Rule

The following components shall never be added twice:

OT Wages

Special Allowance

These components are already included in Gross Pay.

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## SECTION 12: TOTAL CTC CALCULATION

### Formula

```
Total CTC
=
Gross Pay
+
Employer PF
+
Employer PF Admin Charges
+
Employer ESIC
```

---

## SECTION 13: PAYROLL FREEZE RULES

### Payroll Generation Flow

```
Calculate Payroll
↓
Preview Payroll
↓
Generate Payroll
↓
Generate Payslips
↓
Freeze Payroll
```

---

### After Payroll Freeze

Not Allowed:

Modify Payroll

Delete Payroll

Regenerate Payroll

Modify Attendance

---

## Historical Preservation

Previously generated payrolls must remain unchanged permanently.

Future changes to:

Salary Structure

PF %

ESIC %

PT Slabs

OT Rules

must not affect historical payrolls.

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## SECTION 14: INVOICE CALCULATIONS

### Employee Scope

Invoice must include only employees assigned to the selected contract.

Employees assigned to other contracts shall not be included.

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### Invoice Base

Formula:

Invoice Base  
=  
Gross Pay  
+  
Employer PF  
+  
Employer PF Admin Charges  
+  
Employer ESIC

Equivalent to:

Total Company Cost

---

## Service Charge

Formula:

Service Charge  
=  
Invoice Base × Service Charge %

---

## Taxable Amount

Formula:

Taxable Amount  
=  
Invoice Base  
+  
Service Charge

---

## CGST

Formula:



```
CGST
=
Taxable Amount × CGST %
```

---

## SGST

Formula:

```
SGST
=
Taxable Amount × SGST %
```

---

## Final Invoice Amount

Formula:

```
Final Invoice Amount
=
Taxable Amount
+
CGST
+
SGST
```

---

# SECTION 15: INVOICE FREEZE RULES

## Invoice Generation Flow

```
Calculate Invoice
↓
Preview Invoice
↓
Generate Invoice
↓
Generate PDF
```

↓

Freeze Invoice

---

## After Freeze

Not Allowed:

Modify Invoice

Delete Invoice

Regenerate Invoice

---

## Historical Preservation

Changes to:

Service Charge %

CGST %

SGST %

shall apply only to future invoices.

Previously generated invoices remain unchanged permanently.

---

## SECTION 16: ROUNDING RULES

### Database Storage

Store values using:

DECIMAL(12,2)

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## Display Rule

For:

- Payslips
- Payroll Preview
- Invoice Preview
- Reports
- PDFs

Display values rounded to nearest rupee.

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## SECTION 17: SETTINGS CONTROLLED PARAMETERS

The following values must be configurable through Settings:

Default Basic %

Default HRA %

Default Conveyance %

PF %

PF Admin %

ESIC %

Professional Tax Slabs

Maximum OT Hours Per Day

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## SECTION 18: AUDIT REQUIREMENTS

The following actions must be recorded in Audit Logs:

Payroll Generated

Payroll Frozen

Invoice Generated

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Invoice Frozen

Salary Structure Updated

Professional Tax Updated

PF Percentage Updated

ESIC Percentage Updated

Maximum OT Limit Updated

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## FINAL BUSINESS RULE

In case of any conflict between application code and this document:

This Payroll & Invoice Calculation Specification shall be treated as the final source of truth.